Cross-Domain Cooperative Fusion Network for Few-Shot Remote Sensing Imagery Segmentation

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***Abstract*—Few-shot semantic segmentation (FSSS) in remote sensing faces challenges due to limited annotated samples, significant domain shifts, and complex interclass variations. In this article, a cross-domain cooperative fusion network (CDCFNet) is proposed for few-shot remote sensing imagery segmentation across heterogeneous domains. The framework incorporates cross-domain cooperative attention that performs joint channel-spatial attention to achieve dynamic feature alignment between source and target domains. The integration of positional and global–local cooperative attention strengthens domain invariance and enhances the robustness of target recognition. Furthermore, a gated adaptive feature-aligned pyramid transformation (GAF-PTM) is designed to decouple domain-specific and transferable components. This module further employs hierarchical feature weighting and prototype-guided sparse alignment to mitigate overfitting under limited-sample conditions. Comprehensive experiments conducted on multiple cross-domain remote sensing imagery segmentation tasks demonstrate that the proposed method consistently outperforms existing few-shot segmentation baselines in both one-shot and five-shot settings, achieving high accuracy and strong generalization across heterogeneous domains. The experimental results verify the effectiveness and adaptability of the proposed framework.**

***Index Terms*—Cross-domain, domain transfer, few-shot learning, imagery segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing imagery is an end-to-end pixel-wise classification task that assigns each pixel in aerial or satellite images to predefined semantic classes with deep models [1]. As a fundamental geospatial understanding process, this technique converts raw Earth observation data into structured and spatially explicit semantic maps, addressing geospatially heterogeneous targets across large-scale scenes characterized by significant scale variations and complex object topologies [2], [3], [4].

In recent years, with the development of deep learning and the driving force of large-scale data, deep models for remote sensing semantic segmentation have achieved encouraging results [5], [6], [7], [8]. However, deep semantic segmentation models impose demands on large-scale pixel-level annotations, particularly burdensome for high-resolution remote sensing imagery, where expert labeling exceeds several hours/km^2 [3], [9], while struggling with long-tailed class distributions and emerging unseen categories (e.g., novel urban infrastructure or disaster-induced terrain changes) [10], [11]. These difficulties have prompted researchers to focus on few-shot semantic segmentation (FSSS), which learns by meta-transferring knowledge from base classes to novel classes with episodic training, enabling rapid adaptation to novel geospatial targets with merely several support annotation samples [12], [13], [14], [15].

Overall, current FSSS-related research can be summarized into prototype-based and pixel-based paradigms. The prototype-based methods [12], [14], [16], [17], [18] condense support set features of a target class into a compact vector representation (also known as a prototype) via masked average pooling, then segment the query image with pixel-wise similarity matching between query features and this prototype. Conversely, the pixel-based methods [11], [19], [20] bypass intermediate abstraction and directly compute dense feature correlations between all spatial positions in the query and support images with high-dimensional similarity tensors, preserving fine-grained spatial relationships to guide segmentation.

Despite achieving promising performance by integrating prototype-based efficiency and pixel-based granularity, the existing FSSS-related methods for remote sensing imagery are fundamentally limited by the assumption of domain homogeneity [21]. These approaches rely on frozen, source-biased encoders that fail to disentangle domain-invariant semantics from domain-specific interference, such as spectral variations, sensor discrepancies, and geospatial diversity inherent in the data. Consequently, feature distribution misalignment occurs, inducing catastrophic prototype matching failure and significant performance degradation in generalizing to unseen domains.

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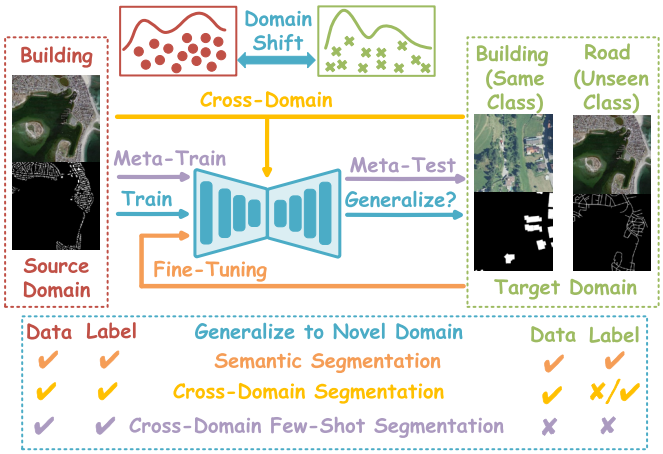


Fig. 1. Different paradigms for diverse semantic segmentation scenarios.

In cross-domain segmentation tasks, the data distributions of the source domain and target domain often exhibit significant distribution discrepancies. As shown in Fig. 1, cross-domain segmentation typically involves training on labeled data from the source domain and then fine-tuning with data from the target domain. However, when only scarce samples are available in the target domain, the model struggles to learn effective information from the target domain. This means the model must accomplish cross-domain segmentation in the target domain without sufficient target-domain supervision. In addition, remote sensing imagery inherently exhibits high intraclass variability and strong interclass similarity due to variations in imaging conditions and scene composition. These characteristics further exacerbate cross-domain distribution shifts, making effective domain transfer particularly challenging [22], [23]. Consequently, directly transferring a model trained on the source domain to the target domain often leads to a sharp decline in segmentation performance due to feature space misalignment.

To address the challenges, this article proposes a cross-domain cooperative fusion network (CDCFNet) for few-shot remote sensing imagery segmentation. The network employs a cross-domain cooperative attention (CDCA) mechanism to enable dynamic feature selection and semantic alignment across domains, alleviating distribution shifts, sample scarcity, and intraclass variations. In addition, a gated adaptive feature-aligned pyramid transformation (GAF-PTM) is introduced to perform multiscale feature fusion and hierarchical semantic alignment, enhancing both multiscale representation capability and domain invariance. By jointly leveraging these components, CDCFNet achieves adaptive alignment between source and target domains, facilitating the extraction of domain-invariant representations and improving robustness under cross-domain and few-shot settings. The major contributions of this article can be summarized as follows.

1) We propose a CDCFNet for few-shot remote sensing image segmentation, which effectively addresses distribution discrepancies and complex feature coupling across domains. By integrating cooperative attention and adaptive feature transformation, the model enables stable and accurate cross-domain knowledge transfer under limited supervision.
2) We design a CDCA that performs dynamic feature alignment through joint channel and spatial attention. By incorporating positional information and global–local context modeling, it enhances domain invariance and robustness, while dynamic feature reweighting mitigates distribution shifts.
3) A GAF-PTM is designed to decouple domain-specific and transferable features through gated adaptation, which maps representations into a category-agnostic and domain-invariant space. In addition, hierarchical feature modeling enables adaptive multilevel feature integration, while prototype-guided sparse alignment alleviates overfitting and improves generalization under few-shot conditions.

The rest of this article is organized as follows. Section II briefly reviews the related work on FSSS and cross-domain few-shot learning. Section III presents the proposed CDCFNet and its key components in detail. Experimental results and analysis are reported in Section IV. The conclusion is drawn in Section V.

II. Related Work

A. Few-Shot Semantic Segmentation

Semantic segmentation is a critical task for achieving pixel-level semantic analysis. The objective is to precisely classify each pixel in an input image into predefined semantic categories, thereby establishing a dense mapping relationship from low-level visual features to high-level semantic understanding [24]. In the condition of extremely scarce labeled samples, FSSS aims to accomplish pixel-level segmentation tasks with more generalized models [25].

According to the modeling paradigm, existing FSSS methods can be classified into the following categories: metric-based methods, parameter-generative methods, attention-based methods, and self-supervision-based methods [26]. Metric-based methods perform pixel-level classification by constructing similarity metrics between support sets and query sets. For example, Shaban et al. [27] first proposed a few-shot segmentation framework that generates classifiers through conditional convolutions. Wang et al. [17] introduced prototype representations to compute feature similarity between support sets and query sets, which enables rapid matching. Parameter-generative methods generate parameters for specific categories from the support set, which can be adapted to the query set. Such methods can rapidly adapt to new categories but rely on the quality of the support samples. In [28], AMP converts support set features into classifier weights through its adaptive parameter generation module. Attention-based approaches introduce channel attention or cross-image attention mechanisms to uncover more effective correlations between the support sets and the query sets. Zhang et al. [10] proposed a CANet that enhances support-query feature interactions through multilayer attention modules. For complex scenarios, Li et al. [29] employed an adaptive subgraph generation module to improve modeling capabilities

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for target structures. Self-supervision-based methods leverage self-supervised learning to achieve stronger representational capability, then rapidly adapt with minimal labeled data. For example, Xian et al. [30] leveraged the contrastive language-image pretraining (CLIP) model for panoptic segmentation and proposed a central index attention mechanism to facilitate knowledge transfer.

B. Cross-Domain Few-Shot Learning

Traditional few-shot learning generally assumes that the training set (source domain) and test set (target domain) have the identical distribution. However, in practical applications, domain shift often occurs due to factors such as different lighting conditions, scene layouts, and object appearances. When there is a substantial distribution shift between the source and target domains, the requirement for strong model generalization becomes more pronounced.

In recent years, researchers have proposed various methods to enhance algorithm performance in cross-domain scenarios. Some researchers have incorporated domain adaptation mechanisms into classical meta-learning methods to enhance cross-domain generalization of the algorithm. For example, Zhao et al. [31] combined prototypical feature alignment with normalized distribution alignment to effectively mitigate differences between support-query feature spaces and distributions. Feature representation learning can capture domain-invariant features and has increasingly been employed to alleviate the impact of domain shift. In [32], a task-dependent adaptive metric is proposed to enhance cross-domain adaptation through task-conditioned embeddings. In addition, to address the impact of insufficient samples and domain differences, some approaches utilize domain adversarial training to augment samples in the target domain or bridge differences between domains. Hu and Ma [33] proposed an adversarial feature augmentation module to reduce the cross-domain feature differences, which can be easily integrated into different methods. Recently, large models have also been applied to cross-domain few-shot learning, subsequently giving rise to a series of transfer learning-based methods [34], [35], [36]. Some studies indicate that pretraining combined with fine-tuning often outperforms meta-learning approaches. For example, Oh et al. [35] found that when pretraining strategies are combined with fine-tuning, it can significantly improve performance under large domain shifts. Zhou et al. [36] proposed an absorption adaptation learning framework that regularizes the fine-tuning procedure. The foundation model extracts and absorbs task-relevant semantic knowledge from the intermediate features of the expert model, which can reduce performance drops from domain gaps and limited annotation.

In semantic segmentation tasks, cross-domain few-shot learning faces more complex challenges due to the requirement for pixel-level annotations. The fundamental difficulty lies in the inability of limited annotated samples to adequately represent the complex distribution of the target domain, while domain shift further aggravates the deterioration of generalization performance. Recent studies have explored various strategies to address cross-domain FSSS [37], [38], [39], [40]. For example, Li et al. [37] proposed matrix-form parameter adapters embedded into backbone features via residual connections, enabling task-specific weight generation. Lu et al. [38] introduced a transductive fine-tuning framework with uncertainty-aware contrastive loss and auxiliary regularization, which effectively enhances support-query alignment under severe domain shifts. Existing studies mainly focus on disentangling domain-specific and domain-invariant features to build shared semantic spaces, often within meta-learning frameworks for episodic training and rapid adaptation. Nevertheless, their robustness remains insufficient under complex domain shifts, such as cross-sensor or cross-resolution remote sensing scenarios.

III. Methodology

A. Overview

Given the scarcity of labeled data and the distribution discrepancy across domains, research in cross-domain few-shot segmentation for remote sensing imagery focuses on building models that are capable of precise segmentation in the target domain. Specifically, the source domain support set is denoted as $S_{source} = \{(x_i^{source}, y_i^{source})\}_{i=1}^{K\times N}$, which contains N classes (N-way) with K labeled samples per class (K-shot). The source domain query set is denoted as $Q_{source} = \{x_q^{source}\}$, which consists of annotated source domain images for model learning. Similarly, the target domain support set is represented as $S_{target} = \{(x_j^{target}, y_j^{target})\}_{j=1}^{K'\times N'}$, where N' is the number of classes in the target domain, and $K' \ll K$ (typically $K' \leq 5$). The target domain query set $Q_{target} = \{x_q^{target}\}$ contains unlabeled target domain images. The objective of the cross-domain few-shot segmentation task is to accurately predict the semantic label $\hat{y}_q^{\text{target}}$ for each pixel in the target query image. This is achieved by minimizing the cross-domain distribution shift and the few-shot segmentation error, which can be formulated as

$$\min_{\theta}\left\{\underbrace{\mathbb{E}_{(x,y)\sim S_{\text{source}}}\mathcal{L}_{\text{seg}}\left(f_\theta(x),y\right)}_{\mathcal{L}_S}+\underbrace{\lambda\cdot\mathcal{D}\left(P_{\text{source}},P_{\text{target}}\right)}_{\mathcal{R}_c}+\underbrace{\gamma\cdot\mathbb{E}_{(x,y)\sim S_{\text{target}}}\mathcal{L}_{\text{seg}}\left(f_\theta(x),y\right)}_{\mathcal{L}_T}\right\} \tag{1}$$

where $\mathcal{L}_S$, $\mathcal{L}_T$, and $\mathcal{R}_c$ represent the source domain supervised loss, target domain few-shot supervised loss, and cross-domain alignment regularization term, respectively. $\mathbb{E}$ denotes the mathematical expectation operator, which averages the loss over all samples in the source domain support set S_{source}. In addition, $f_\theta(x)$ denotes the segmentation model with parameters θ, and it takes an image x as input and outputs a pixel-level prediction. $\mathcal{L}_{seg}$ stands for the loss function for the segmentation task (e.g., cross-entropy loss), which measures the discrepancy between the prediction and the ground truth label. λ is the regularization coefficient that controls the weight of the cross-domain alignment term. $\mathcal{D}$ is the domain discrepancy metric function, used to quantify the data distribution

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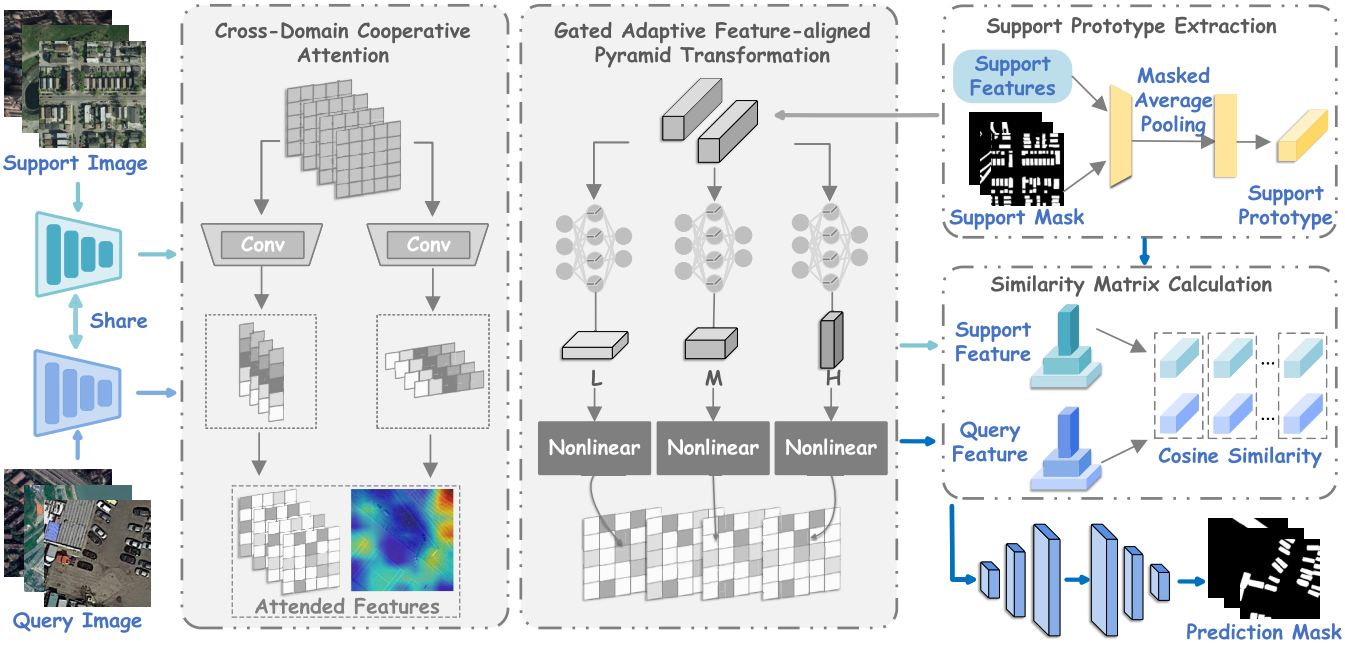


Fig. 2. Specific network architecture of CDCFNet. This framework primarily consists of two components: 1) the CDCA for adaptive cross-domain recalibration, as shown in the purple frame; and 2) the GAF-PTM for hierarchical feature alignment, as shown in the blue frame. This figure illustrates the data flow from feature extraction to cross-domain alignment and hierarchical matching, highlighting the interactions between different components.

difference between the source and target domains. P_{source} and P_{target} denote the data distributions of the source and target domains, respectively. γ is a hyperparameter that adjusts the weight of the target domain supervised loss. In remote sensing scenarios, the scarcity of samples, high intraclass variability, and strong interclass similarity all add to the difficulty of semantic segmentation tasks. Under cross-domain conditions, the significant differences in data distribution further raise the requirement for the generalization performance of models.

Therefore, this article proposes a cross-domain few-shot segmentation framework designed for remote sensing scenarios characterized by domain shifts and limited annotations. By decoupling domain-specific and domain-invariant features, this framework constructs a semantic space that is shared across domains. Simultaneously, it incorporates a meta-learning paradigm to simulate the episodic training mechanism under cross-domain few-shot scenarios, which enhances the adaptation capability of the model. Unlike traditional domain adaptation methods that require access to target-domain data during training, the proposed framework is trained exclusively on a few labeled source-domain samples, without any target-domain involvement in the global training stage. During inference, the model leverages only a small number of labeled target-domain support samples to guide segmentation on new target images, enabling rapid adaptation under data-scarce conditions. By disentangling domain-specific and domain-invariant representations and enforcing semantic correspondence, the framework enables training on source data and adaptation through support data, which bridges the gap between domain generalization and few-shot segmentation in remote sensing.

B. Cross-Domain Cooperative Fusion Network

Achieving effective cross-domain knowledge transfer is particularly challenging due to the evident disparity in feature distributions between the source and target domains. Consequently, cross-domain few-shot learning models must implement adaptive mechanisms to align features across domains as well as extract domain-invariant high-level semantic features to overcome the above challenges. The specific network architecture of CDCFNet is illustrated in Fig. 2, which consists of two main components: the cross-domain cooperative attention and the GAF-PTM. Given a support image and a query image, a shared backbone network is first employed to extract multilevel feature representations. These features are then processed through two successive stages: cross-domain feature alignment and hierarchical feature transformation.

Specifically, the CDCA module integrates spatial location information through channel-wise adaptive normalization and performs joint spatial-channel modeling to enable dynamic feature selection. By capturing both global context and local details across multiple feature levels, CDCA enhances feature consistency between domains and improves robustness under diverse appearance conditions. Subsequently, the GAF-PTM module performs hierarchical feature transformation. It employs a dynamic gating mechanism to adaptively fuse linearly transformed features with residual representations, thereby suppressing domain-related and category-related noise. Meanwhile, through multiscale feature integration, the module aligns feature distributions and enhances both perceptual capability and domain invariance.

By combining CDCA and GAF-PTM, the proposed framework effectively bridges domain discrepancies while integrating local details with global semantic information. The

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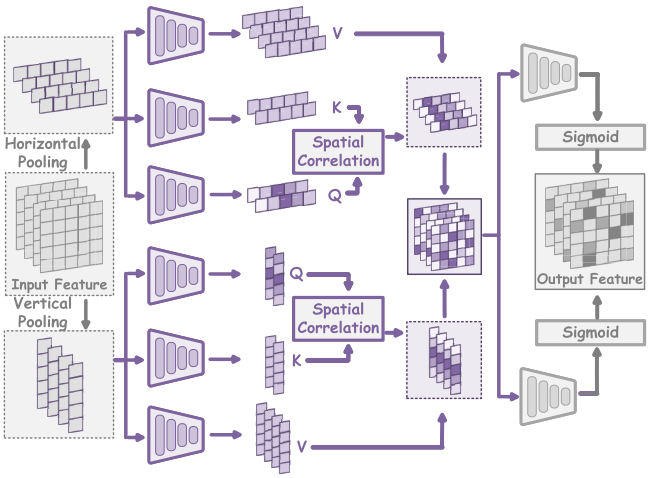


Fig. 3. Cross-domain cooperative attention.

transformed features are then fed into a feature encoder–decoder module. In this article, the module is constructed with a 2-D convolutional context decoder and a 4-D convolutional pyramid encoder, which can efficiently generate prediction masks in a coarse-to-fine manner.

C. Cross-Domain Cooperative Attention

The CDCA module aims to leverage spatial and contextual information while improving robustness to scale variations. In addition, through dimension decoupling and local enhancement strategies, it balances efficiency and accuracy. As shown in Fig. 3, the CDCA module employs a cooperative attention mechanism across channel and spatial dimensions to dynamically adjust feature weights. This enables automatic alignment and position-aware adaptation of cross-domain features, effectively mitigating the impact of domain shifts and improving the adaptability to remote sensing imagery from different regions. Simultaneously, by integrating global and local attention, the module captures structural characteristics of targets rather than relying on superficial textures, which strengthens robustness to appearance variations.

Specifically, CDCA decomposes the traditional 2-D spatial modeling process into independent encoding streams along the horizontal axis (H, 1) and vertical axis (1, W), which capture information for each channel separately in these two directions. The output of the cth channel at height h can be formulated as

$$\mathcal{P}_c^h(h) = \frac{1}{W} \sum_{0 \le i \le W} x_c(h, i). \tag{2}$$

Similarly, the output of the cth channel with width w can be written as

$$\mathcal{P}_c^w(w) = \frac{1}{H} \sum_{0 \le j \le H} x_c(j, w). \tag{3}$$

This decomposition strategy effectively captures long-range spatial dependencies while preserving local positional information, which provides more granular feature representations for subsequent attention computations. The orthogonal dual-channel spatial information encoding stream is constructed through the above two transformations, which differs from the global feature compression strategy employed by traditional channel attention mechanisms. Classical methods decrease spatial dimensions across the entire space, which may lead to the loss of pixel-level coordinate information. The dual encoding streams generate complementary effects, which focus on establishing cross-regional semantic associations in the horizontal stream while reinforcing local structural constraints in the vertical stream. The collaboration between the two streams effectively balances the dual requirements of global context understanding and pixel-level coordinate perception.

A two-stage feature projection will be performed on the obtained 1-D features. Taking the horizontal-axis feature vector as an example, for the feature tensor $\mathbf{P} \in \mathbb{R}^{C \times 1 \times H}$, channel compression is first achieved through a cascaded 1×1 convolution kernel, obtaining the reduced-dimensional query tensor $\mathbf{Q} \in \mathbb{R}^{C' \times 1 \times H}$ and key tensor $\mathbf{K} \in \mathbb{R}^{C' \times 1 \times H}$. The compression rate is set to $C' = \lfloor C/4 \rfloor$ to balance information retention and computational efficiency. This projection process can be formalized as

$$\mathbf{Q} = \mathcal{D}_q * \mathbf{P}, \quad \mathbf{K} = \mathcal{D}_k * \mathbf{P} \tag{4}$$

where $\mathcal{D}$ denotes the learnable projection matrix, and $*$ indicates the convolution operation. Subsequently, the axial spatial correlation matrix $\mathbf{M} \in \mathbb{R}^{H \times H}$ is constructed through similarity measurement and probability normalization. The computation can be formulated as

$$\begin{aligned} g_{i,e} &= \mathbf{Q}_e \mathbf{K}_i^\top \\ m_{i,e} &= \frac{\exp(g_{i,e})}{\sum_{j=1}^{H} \exp(g_{j,e})} \end{aligned} \tag{5}$$

where $\mathbf{Q}_e \in \mathbb{R}^{C'}$ denotes the query vector at any spatial location $e \in \{1, \ldots, H\}$. $g_{i,e}$ represents the correlation metric, and $m_{i,e}$ is the normalized attention weight at position (i, e). By introducing a value projection tensor $\mathbf{V} = \mathcal{D}_v * \mathbf{P} \in \mathbb{R}^{C \times 1 \times H}$, where $\mathcal{D}_v \in \mathbb{R}^{C \times C}$ is a learnable transformation, contextual feature aggregation is further performed. The process is computed as

$$\mathbf{P}'_e = \sum_{i=1}^{H} m_{i,e} \mathbf{V}_i + \mathbf{P}_e \tag{6}$$

where $\mathbf{P}'_e$ is the feature vector at position e.

To further accentuate the target regions precisely, the CDCA module refines and aggregates the obtained features via a position attention-generated strategy. This process is formally defined as follows:

$$z = \delta\left(F\left(\left[\mathbf{P}'_h, \mathbf{P}'_w\right]\right)\right) \tag{7}$$

where $[\cdot, \cdot]$ denotes the concatenation operation, and $F(\cdot)$ is the convolutional transformation function. $\delta(\cdot)$ is a nonlinear activation function. The resulting $\mathbf{z} \in \mathbb{R}^{C/r \times (H+W)}$ is an intermediate feature tensor, which is then split along the horizontal and vertical axes into $\mathbf{z}_h \in \mathbb{R}^{C/r \times H}$ and $\mathbf{z}_w \in \mathbb{R}^{C/r \times W}$, respectively. Subsequently, $\mathbf{z}_h$ and $\mathbf{z}_w$ are fed into convolutional transformation functions followed by nonlinear activation, which can be expressed as:

$$l^h = \sigma\left(F_h\left(z^h\right)\right), l^w = \sigma\left(F_w\left(z^w\right)\right). \tag{8}$$

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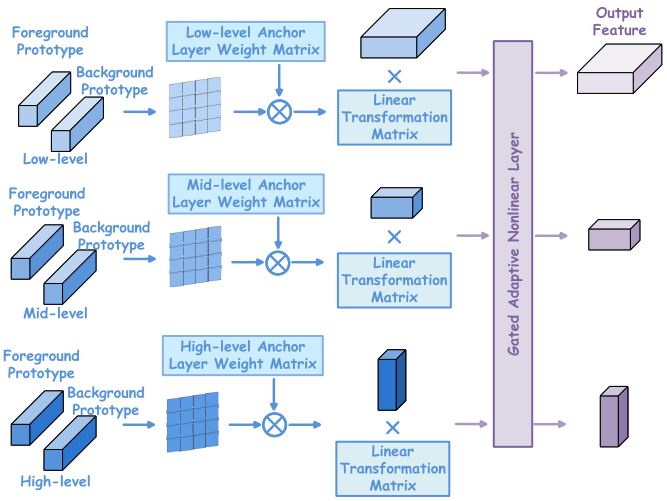


Fig. 4. GAF-PTM.

The transformed outputs l_h and l_w have matched the channel dimension of the original input. The final refined feature map $\mathbf{Y}$ can be obtained as

$$y_c(i,j) = x_c(i,j) \times l_c^h(i) \times l_c^w((j)). \tag{9}$$

On one hand, CDCA extracts features simultaneously from multiple dimensions, including spatial, channel, and coordinate information, which achieves cooperative integration of multidimensional features. On the other hand, by integrating attention mechanisms that capture long-range dependencies and local details, each pixel can perceive both global semantic context and surrounding local feature information, enabling the collaboration of global and local representations. This design further allows the model to effectively handle objects with extreme scale variations, as global dependencies facilitate the modeling of large-scale structures, while local details enhance the representation of small-scale targets.

D. Gated Adaptive Feature-Aligned Pyramid Transformation

GAF-PTM module transforms category-specific features into category-agnostic representations while narrowing the domain gap, thereby identifying common characteristics across remote sensing imagery and achieving domain feature alignment. Since the resulting category-agnostic metric space is constant, the downstream segmentation task can perform prediction more effectively in this stable space.

Specifically, GAF-PTM facilitates cross-category knowledge transfer through feature space decoupling and reconstruction. It can be decomposed into three key components.

1) *Feature Hierarchization:* Decomposing deep features into *low-level*, *mid-level*, and *high-level* representations.
2) *Spatial Projection:* Projecting the hierarchical features into a category-agnostic metric space via linear transformation.
3) *Gated Fusion:* Dynamically integrating the transformed features with original features to preserve discriminative information.

The mathematical model of GAF-PTM can be expressed as follows:

$$\tilde{\mathcal{F}} = \mathcal{T}(\mathcal{F}cls) \oplus \mathcal{G}(\mathcal{F}dom) \tag{10}$$

where $\mathcal{T}$ denotes a linear transformation. $\mathcal{G}$ is a gating function, and $\oplus$ represents an adaptive fusion operation. To minimize the number of introduced learnable parameters, GAF-PTM employs a transformation matrix as a mapping operator. As shown in Fig. 4, the transformation matrix $\mathbf{U}$ is computed using the weight matrix $\mathbf{A}$ from the anchor layer and the prototype matrix $\mathbf{D}$ derived from support images. Specifically, in a one-way one-shot task, $\{\mathbf{X}_s^l\}_{l=1}^L$ denotes the set of intermediate feature maps from the support image at L layers. The foreground prototype for each feature map $\mathbf{X}_s^l \in \mathbb{R}^{C_l \times H_l \times W_l}$ can be computed using the support mask $\mathbf{M}_s \in \{0,1\}^{H \times W}$ via masked average pooling

$$\mathbf{D}_{f,l}^s = \frac{\sum_i \mathbf{X}_{l,i}^s \zeta_l(\mathbf{M}^s)_i}{\sum_i \zeta_l(\mathbf{M}^s)_i} \tag{11}$$

where $\mathbf{D}_{f,l}^s \in \mathbb{R}^{C_l}$, and i indexes the 2-D spatial positions in the feature map. $\zeta_l(\cdot)$ denotes a bilinear interpolation function that expands the input tensor to the spatial dimensions $\mathbb{R}^{C_l \times H_l \times W_l}$ of the lth layer feature map. Similarly, the background prototype $\mathbf{D}_{b,l}^s$ can be computed in the same manner. Therefore, the prototype matrix $\mathbf{D}_l^s$ can be calculated as

$$\mathbf{D}_l^s = \left[\frac{\mathbf{D}_{f,l}^s}{\|\mathbf{D}_{f,l}^s\|}, \frac{\mathbf{D}_{b,l}^s}{\|\mathbf{D}_{b,l}^s\|} \right]. \tag{12}$$

Correspondingly, the anchor weight matrix $\mathbf{A}^l$ is calculated as

$$\mathbf{A}_{\mathbf{i,l}} = \frac{\mathbf{A}_{f,l}}{\|\mathbf{A}_{f,l}\|}, \frac{\mathbf{A}_{b,l}}{\|\mathbf{A}_{b,l}\|}. \tag{13}$$

In general, $\mathbf{D}_l^s$ is a nonsquare matrix, and its generalized inverse can be computed as

$$\mathbf{D}_l^{s+} = \left(\mathbf{D}_l^{sT}\mathbf{D}_l^s\right)^{-1}\mathbf{D}_l^{sT}. \tag{14}$$

Therefore, the transformation matrix can be derived as

$$\mathbf{U}_l = \mathbf{A}_l\mathbf{D}_l^{s+}. \tag{15}$$

To enhance representational capacity, this work introduces a gated adaptive nonlinear layer designed to dynamically calibrate feature transformation strength, thereby ensuring the preservation of critical semantic integrity. Specifically, the hierarchical feature weighting is realized through a learnable gating mechanism, where the weighting coefficients are generated in a data-dependent manner for each feature level. The mathematical model can be expressed as follows:

$$\mathcal{G}_l(\mathbf{X}) = \sigma(\Phi_l\mathbf{X} + \mathbf{b}_l) \odot (\mathbf{U}_l\mathbf{X}) + (1 - \sigma(\Phi_l\mathbf{X} + \mathbf{b}_l)) \odot \mathbf{X} \tag{16}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function. Φ_l is a learnable parameter matrix. $\odot$ represents the Hadamard product for channel-wise weighting, and b_l is the bias term that adjusts the activation threshold of the gating weights. The term $\sigma(\Phi_l\mathbf{X} + \mathbf{b}_l)$ serves as the adaptive weighting coefficient, which is learned end-to-end via backpropagation and controls the contribution of the transformed features and the original features. The gating mechanism dynamically fuses

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the linear-transformed results with the original features, while the residual connection ensures stable gradient propagation. In addition, independent gating parameters are employed at different feature levels, enabling hierarchical feature weighting across low-, mid-, and high-level representations.

For the query feature $\mathbf{X}_l^q$ and the support feature $\mathbf{X}_l^s$, the gated enhancement transformation is separately applied as follows:

$$\begin{aligned}\tilde{\mathbf{X}}_l^q &= \mathcal{G}_l\left(\mathbf{X}_l^q\right)\\ \tilde{\mathbf{X}}_l^s &= \mathcal{G}_l\left(\mathbf{X}_l^s \odot \bar{\zeta}_l\left(\mathbf{M}^s\right)\right).\end{aligned} \tag{17}$$

This sparsity is implicitly achieved through the masking operation and prototype-guided feature modulation in (17). Specifically, instead of establishing dense correspondences between all spatial locations, the alignment process is constrained by prototype consistency, where only regions that are semantically relevant to the foreground or background prototypes are emphasized. This effectively suppresses noisy or irrelevant feature interactions and reduces the risk of overfitting to limited support samples. Feature matching is then achieved by constructing a 4-D hyper-correlation tensor, where the similarity between query and support features is measured using cosine similarity, as shown

$$\mathbf{C}_l(i,j) = \mathrm{ReLU}\left(\frac{\left\langle \tilde{\mathbf{X}}_l^q(i), \tilde{\mathbf{X}}_l^s(j)\right\rangle}{\left\|\tilde{\mathbf{X}}_l^q(i)\right\|_2 \left\|\tilde{\mathbf{X}}_l^s(j)\right\|_2}\right) \tag{18}$$

where i and j denote the spatial position indices of the query and support features, respectively, and $\langle\cdot,\cdot\rangle$ denotes the inner product operation. The normalization term corresponds to L_2 normalization, resulting in a cosine similarity that quantifies the semantic correlation between spatial locations. The hierarchical correlation tensor $\{C_l\}_{l=1}^L$ is concatenated and subsequently passed as input to the segmentation decoder.

To balance model capacity and generalization performance, we share three independent anchor layers across different level features. Meanwhile, the gating parameters Φ_l are learned independently at each level, and the residual connection dimensions are aligned using the 1×1 convolution. The proposed GAF-PTM integrates low-level, mid-level, and high-level features for multigranularity information complementation, which achieves hierarchical cooperation of multiscale features and further complements the CDCA module in handling scale variations. By coordinating the dynamic components (gating networks) with static components (pyramid structure), the framework achieves a cooperation between adaptivity and stability.

IV. Experimental Studies

A. Datasets

To demonstrate the effectiveness of our proposed method, five datasets are employed in the experiments: DLRSD-4^i [19], iSAID-5^i [41], iSAID-4^i [19], Inria (building) [42], and CHN6-CUG (road) [43].

1) *DLRSD-4^i*: This dataset is a few-shot aerial image segmentation benchmark constructed from the DLRSD dataset [44], [45]. The original DLRSD dataset contains 7002 high-resolution remote sensing images with a size of 256×256 pixels, covering 17 semantic categories. To enhance spatial heterogeneity and contextual diversity, the field category is removed since it is characterized by homogeneous foreground textures and offers limited discriminative background information. The remaining 16 object categories constitute the experimental dataset. A hierarchical sampling strategy is adopted to divide the 16 classes evenly into four mutually exclusive subsets. Following a fourfold evaluation paradigm, three subsets are used for training and one for validation in each round, which ensures independence across class distributions. The upper part of Fig. 5 displays sample images from these 16 categories along with their corresponding segmentation masks. This dataset is characterized by complex scenes, limited samples, and significant inter-domain variations, which is suitable for evaluating the few-shot generalization capability of models.
2) *iSAID-5^i*: Yao et al. [41] reconstructed the aerial instance segmentation dataset iSAID [46] and released the iSAID-5^i dataset. It contains 2806 high-resolution aerial images with 655 451 object instances across 15 categories, notable for its large scale and fine-grained object diversity. This dataset is widely used to effectively test the capacity of models for fine-grained discrimination under low-sample conditions.
3) *iSAID-4^i*: This dataset is designed to establish a larger scale and more semantically diverse few-shot segmentation benchmark. It is built upon the iSAID-5^i aerial imagery collection [41], further enhanced by integrating urban building annotations from the Massachusetts Building Dataset [47] to form an augmented dataset with 16 land-cover categories. As shown in Fig. 5, iSAID-4^i consists of 28 214 high-resolution images of 256×256 pixels and follows a strict fourfold cross-validation protocol. Featuring cross-modal fusion, complex scenes, and high semantic diversity, iSAID-4^i provides a rigorous benchmark for assessing cross-domain adaptation and semantic understanding in complex urban landscapes.
4) *Inria (Building):* As shown in Fig. 6, this dataset is a high-resolution remote sensing collection designed for urban building detection and semantic segmentation. It provides binary annotations, where pixels are categorized as either *building* or *nonbuilding*. In the experiments, nonbuilding regions are uniformly treated as background classes to align with the segmentation task.
5) *CHN6-CUG (Road):* This dataset is a road extraction benchmark created by Zhu et al. [43] for remote sensing research, which is illustrated in Fig. 6. It contains satellite imagery covering road networks across six major cities in China. The dataset consists of 4511 annotated images with a spatial size of 512×512 pixels and a spatial resolution of 50 cm/pixel, which enables rigorous evaluation of the generalization across diverse geographic and morphological road patterns.

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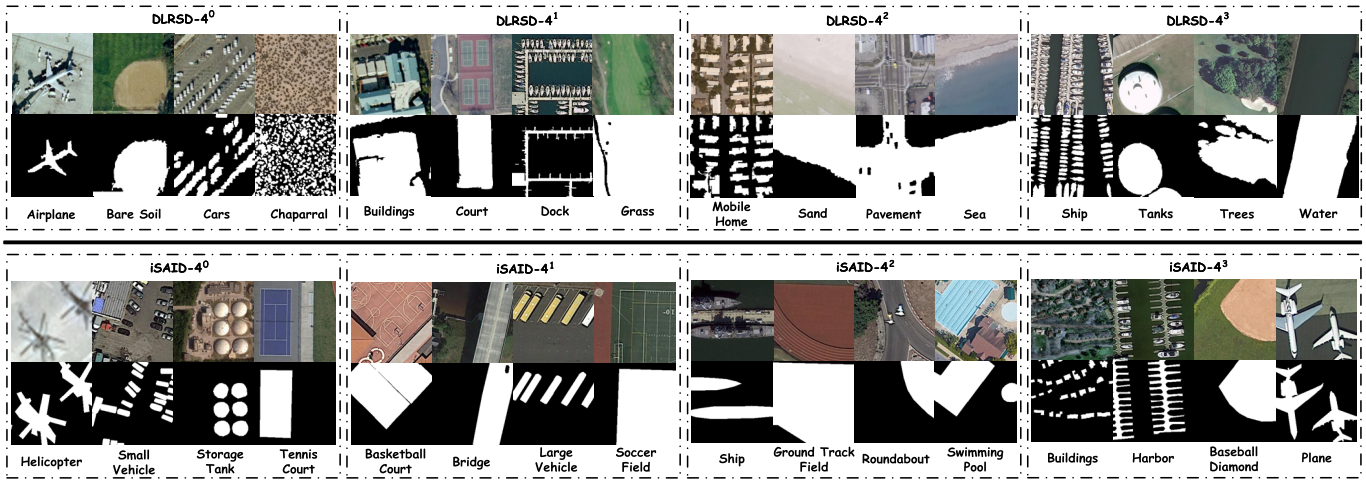


Fig. 5. Cross-domain few-shot segmentation datasets of DLRSD-4^i and iSAID-4^i.

TABLE I
DATASET CONFIGURATION FOR DIFFERENT FEW-SHOT SEGMENTATION EXPERIMENTS

Experiments	Dataset
Few-Shot Segmentation	iSAID-5^i [41]
Cross-Domain Few-Shot Segmentation	[Multi-Category Cross-Domain Few-Shot Segmentation] (1) DLRSD-4^i→iSAID-4^i [19]; (2) iSAID-4^i→DLRSD-4^i [19]
	[Single-Category Cross-Domain Few-Shot Segmentation] (1) Inria (Building) [42]→CHN6-CUG (Road) [43]; (2) CHN6-CUG (Road) [43]→Inria (Building) [42]
	[Heterogeneous-Category Cross-Domain Few-Shot Segmentation] Inria (Building) [42]→iSAID-5^i [41]

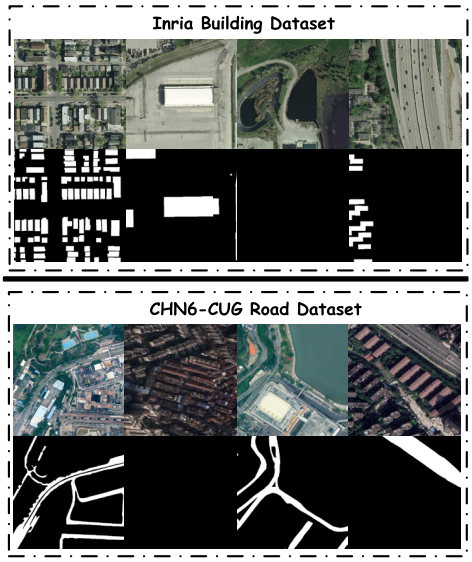


Fig. 6. Cross-domain few-shot segmentation datasets of Inria (building) and CHN6-CUG (road).

These datasets are designed to validate the effectiveness of different methods in both few-shot segmentation and cross-domain few-shot segmentation tasks. Accordingly, depending on the experimental objectives, the aforementioned datasets are divided into the following two categories, with detailed configurations provided in Table I.

1) Few-Shot Segmentation Datasets: To evaluate the segmentation performance in in-domain few-shot scenarios, the iSAID-5^i dataset is adopted as the benchmark. As mentioned before, this dataset covers a wide range of remote sensing scene categories and is constructed following a strict fivefold splitting strategy, ensuring diversity in task partitioning and categorical balance.

2) Cross-Domain Few-Shot Segmentation Datasets: To examine the adaptability of different methods in cross-domain settings, we construct multiple testing scenarios by combining different datasets. Fig. 7 further presents the t-SNE feature visualization of different cross-domain datasets. As observed from the figure, there exists significant distribution overlap between DLRSD-4^i and iSAID-4^i. On such datasets, the model can leverage this distribution overlap to achieve smoother domain adaptation and knowledge transfer, thereby demonstrating stronger cross-domain generalization capability. In contrast, the t-SNE feature clusters of CHN6-CUG and Inria, as well as Inria and iSAID-5^i, are almost completely separated. This characteristic can be utilized to construct challenging extreme domain shift scenarios, which can validate the robustness and adaptive capacity of models under conditions of significant distribution discrepancy and substantial semantic gap. Based on the distribution characteristics of the data, different cross-domain few-shot segmentation datasets are categorized as follows.

1) *Multicategory Cross-Domain Few-Shot Segmentation:* Models are trained on multiple source domain data

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TABLE II

OVERALL PERFORMANCE (IN %) COMPARISON ON THE FEW-SHOT REMOTE SENSING IMAGERY SEGMENTATION BENCHMARK iSAID-5^i DATASET, THE BEST RESULT IS INDICATED IN BOLD, WHERE THE BACKBONE IS "RESNET-50"

Method	1-Shot					5-Shot				
	Fold-0	Fold-1	Fold-2	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	**mIoU**	**FB-IoU**
CANet [10]	25.51	13.50	24.45	21.15	56.64	29.32	21.85	26.91	26.03	59.4 6
PANet [17]	27.56	17.23	24.60	23.13	56.56	36.54	16.05	26.22	26.27	57.37
PFENet [16]	35.84	23.35	27.20	28.80	60.09	42.42	25.34	33.00	33.59	63.25
HSNet [11]	28.49	24.66	**39.36**	30.84	54.72	36.64	**29.87**	40.11	35.54	57.49
SCL [48]	34.78	22.77	31.20	29.58	61.30	41.29	25.73	37.70	34.91	64.13
BAM [12]	39.43	21.69	28.64	29.92	62.04	43.29	27.92	38.62	36.61	65.00
DCP [49]	37.83	22.86	28.92	29.87	62.36	41.52	28.18	33.43	34.38	63.77
DML [50]	32.96	18.98	26.27	26.07	58.93	33.58	22.05	29.77	28.47	59.23
DMML-Net [14]	28.45	21.02	23.46	24.31	57.78	30.61	23.85	24.08	26.18	58.26
NERTNet [51]	34.93	23.95	28.56	29.15	59.97	44.83	26.73	37.19	36.25	64.45
SDM [41]	27.96	21.99	27.82	25.92	59.58	28.50	25.23	31.07	28.27	59.90
TBPN [52]	29.33	16.84	25.47	23.88	57.34	30.98	20.42	28.07	26.49	29.67
PCNet [53]	40.24	24.64	31.31	32.06	62.64	45.31	28.19	37.36	36.95	65.12
R^2Net [18]	41.22	21.64	35.28	32.71	63.86	46.45	25.80	39.84	37.36	66.18
MS^2A^2Net [19]	34.96	25.30	30.71	30.32	59.81	42.30	27.10	36.31	35.24	62.01
SD-AANet [54]	30.52	**29.31**	18.46	26.10	–	28.01	21.38	16.79	22.06	–
CDCFNet	**41.83**	22.35	37.17	**33.78**	**64.43**	**47.02**	27.13	**41.32**	**38.49**	**66.74**

and transferred to different but semantically related categories in the target domain to evaluate their cross-category generalization capability.

2) *Single-Category Cross-Domain Few-Shot Segmentation:* The model performs cross-domain transfer within a single semantic category to examine feature invariance and generalization performance when encountering the same category but different domain distributions.
3) *Heterogeneous-Category Cross-Domain Few-Shot Segmentation:* Transfer is conducted between source and target domains with completely different semantic categories, which aims to assess the robustness and adaptability of models under conditions of extreme domain and semantic dissimilarity.

B. Implementation Details

To ensure scientific comparability of experimental results, this article uniformly employs a single NVIDIA GeForce RTX 4090, with underlying acceleration support provided by the CUDA parallel computing architecture (version 11.6). The CDCFNet model adopts the following optimization configuration: stochastic gradient descent (SGD) is employed as the optimizer, with a batch size of 12 and a training schedule of 300 epochs. The initial learning rate is set to 2×10^{-3}, and the weight decay coefficient for regularization is 1×10^{-6}. The momentum remains fixed at 0.9. The computational cost mainly arises from the CDCA module and the hyper-correlation computation in GAF-PTM. The CDCA module introduces linear complexity with respect to the feature size, while the hyper-correlation is computed on feature maps, making the overall computation manageable in practice.

ResNet-50 and ResNet-101 are employed as backbone architectures throughout all experiments. These backbones help validate the universality and architectural flexibility of the proposed method. To ensure fairness, all comparative experiments are conducted using publicly available code and specified execution protocols, with results tuned to achieve optimal performance for evaluation.

Two widely adopted evaluation metrics in few-shot segmentation are employed: mean intersection over union (mIoU) and foreground–background intersection over union (FB-IoU). The mIoU metric is calculated as the arithmetic mean of IoU values across all folds, which can provide a fine-grained assessment of segmentation accuracy. By evaluating the inter-class discrimination capability for different target categories, mIoU also comprehensively reflects the overall performance and generalization ability of the model. In contrast, FB-IoU adopts a simplified evaluation strategy that disregards specific object categories and performs binary classification statistics only on foreground and background regions.

C. Comparison With State-of-the-Art

1) Comparison Experiments in Few-Shot Aerial Image Segmentation: To validate the effectiveness of the proposed method, a variety of few-shot segmentation methods were selected for comparison, namely, CANet [10], PANet [17], PFENet [16], HSNet [11], SCL [48], BAM [12], DCP [49], DML [50], DMML-Net [14], NERTNet [51], SDM [41], TBPN [52], PCNet [53], R^2Net [18], MS^2A^2Net [19], and SD-AANet [54].

Table II presents the comparative experimental results of different few-shot segmentation algorithms on the iSAID-5^i dataset using ResNet-50 as the backbone network. CDCFNet achieves superior performance in both mIoU and FB-IoU metrics, which can be attributed to its effective feature alignment and enhanced discriminative representation learning. Under the five-shot setting, the performance improvement of CDCFNet becomes more pronounced, which validates that the proposed architecture can better leverage additional support samples to refine feature representations and strengthen domain-invariant characteristics.

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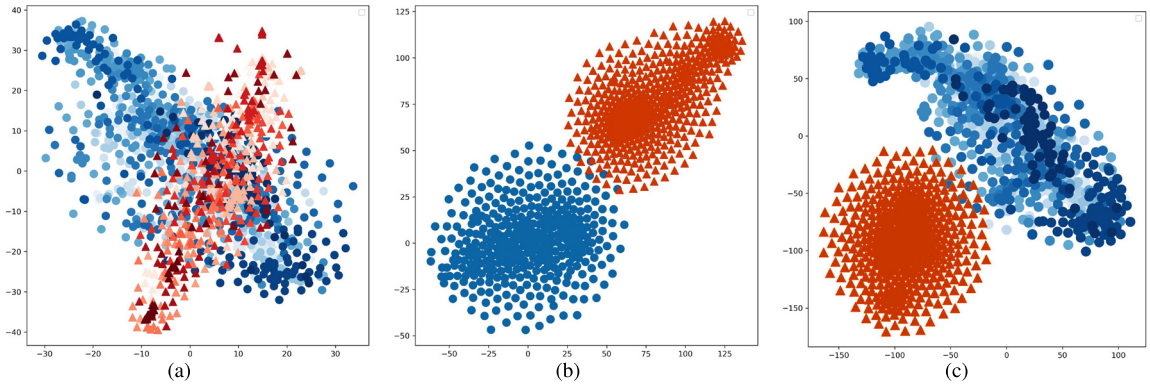


Fig. 7. Comparison of t-SNE feature visualization from different cross-domain datasets. (a) DLRSD-4^i (red) and iSAID-4^i (blue) datasets. (b) CHN6-CUG (red) and Inria (blue) datasets. (c) Inria (red) and iSAID-5^i (blue) datasets.

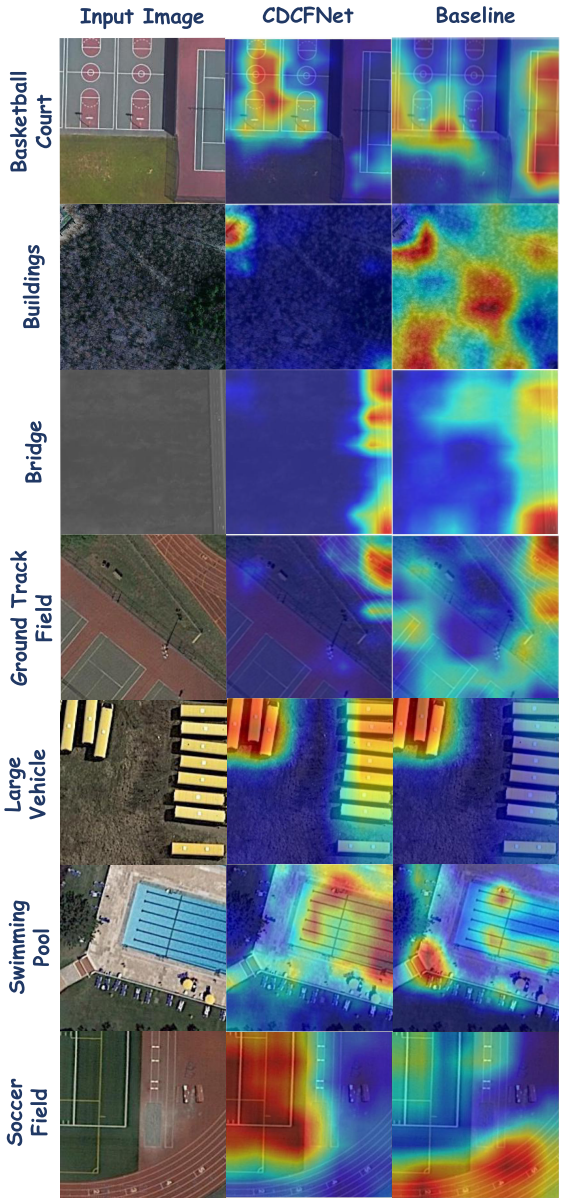


Fig. 8. Heatmap comparisons between CDCFNet and the baseline.

Fig. 8 presents qualitative heatmap comparisons between the proposed CDCFNet and the baseline model under the same backbone network. CDCFNet exhibits more concentrated and accurate attention on target regions, while the baseline shows scattered or misaligned responses. For example, in the basketball court and soccer field scenarios, CDCFNet focuses on the correct regions while suppressing irrelevant background, whereas the baseline tends to attend to surrounding structures. Similar improvements are observed in the swimming pool and large vehicle cases, where CDCFNet produces more complete and consistent activations, highlighting its superior semantic awareness and feature localization capability.

Fig. 9 further illustrates segmentation results under both one-shot and five-shot settings. Even with a single support sample, CDCFNet achieves accurate segmentation, demonstrating strong generalization ability, especially for partially visible targets such as basketball courts and airplanes. With five support samples, the segmentation performance is further improved. For instance, in cases where only small portions of ships or baseball fields are visible, the model can still accurately infer and segment the complete target regions, indicating robust local semantic reasoning under few-shot conditions.

2) Comparison Experiments in Cross-Domain Few-Shot Aerial Image Segmentation: In this section, experiments are conducted on the multicategory cross-domain few-shot segmentation dataset, and several methods (PFENet [16], VAT [55], HSNet [11], BAM [12], HDMNet [56], R^2Net [18], and PATNet [21]) are selected for comparison.

Table II presents the comparative experimental results of different few-shot segmentation algorithms using ResNet-50 and ResNet-101 as backbone networks under one-shot and five-shot conditions. It can be seen that CDCFNet achieves superior results in most scenarios. Under the one-shot setting with ResNet-50 as the backbone, CDCFNet achieves an mIoU improvement of 1.08 over the next-optimal method. When a deeper ResNet-101 backbone is employed, the mIoU gain further increases to 1.71, which suggests that deeper architectures enhance the discriminative power of feature representations. In the five-shot setting, CDCFNet attains performance improvements of 1.14 and 2.19 over the next-optimal methods with ResNet-50 and ResNet-101, respectively. Both results exceed the gains under the one-shot scenario. A similar trend is also observed in the cross-domain experiments from iSAID-4^i to DLRSD-4^i, which validates the effectiveness of the

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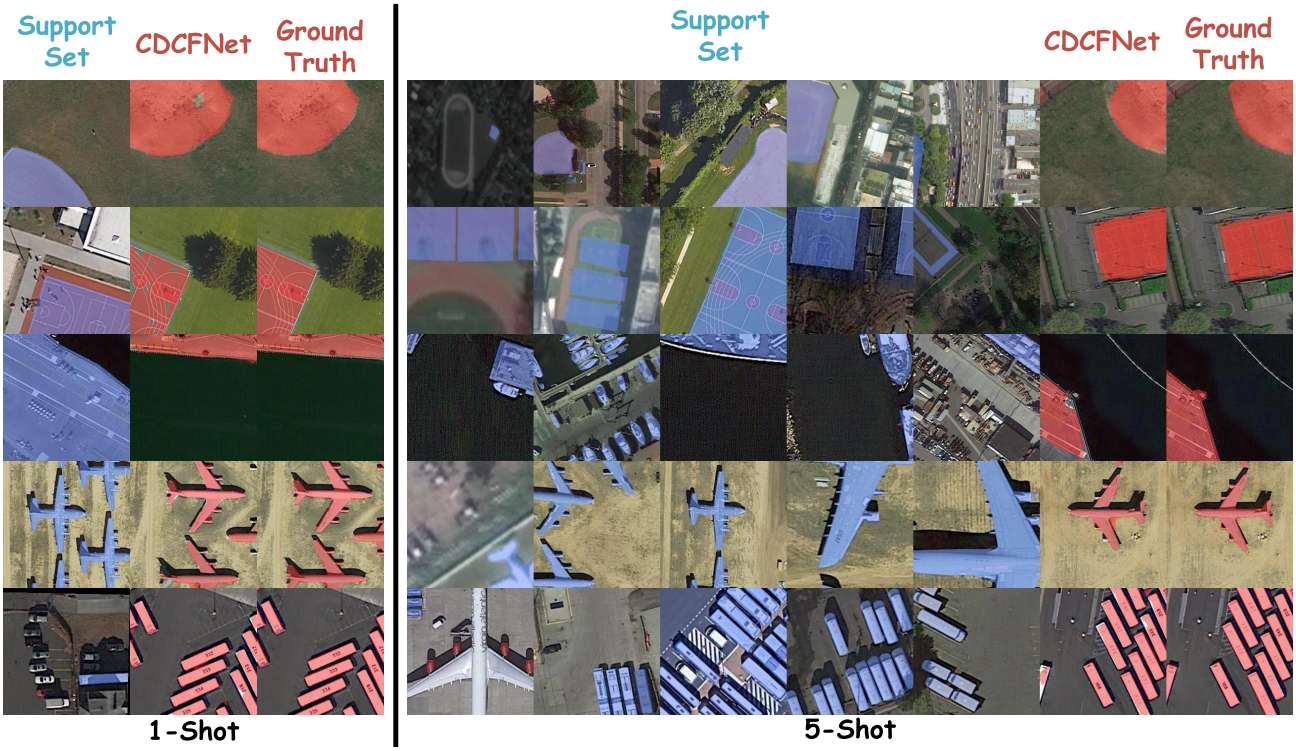


Fig. 9. Segmentation results of CDCFNet on iSAID-5^i dataset.

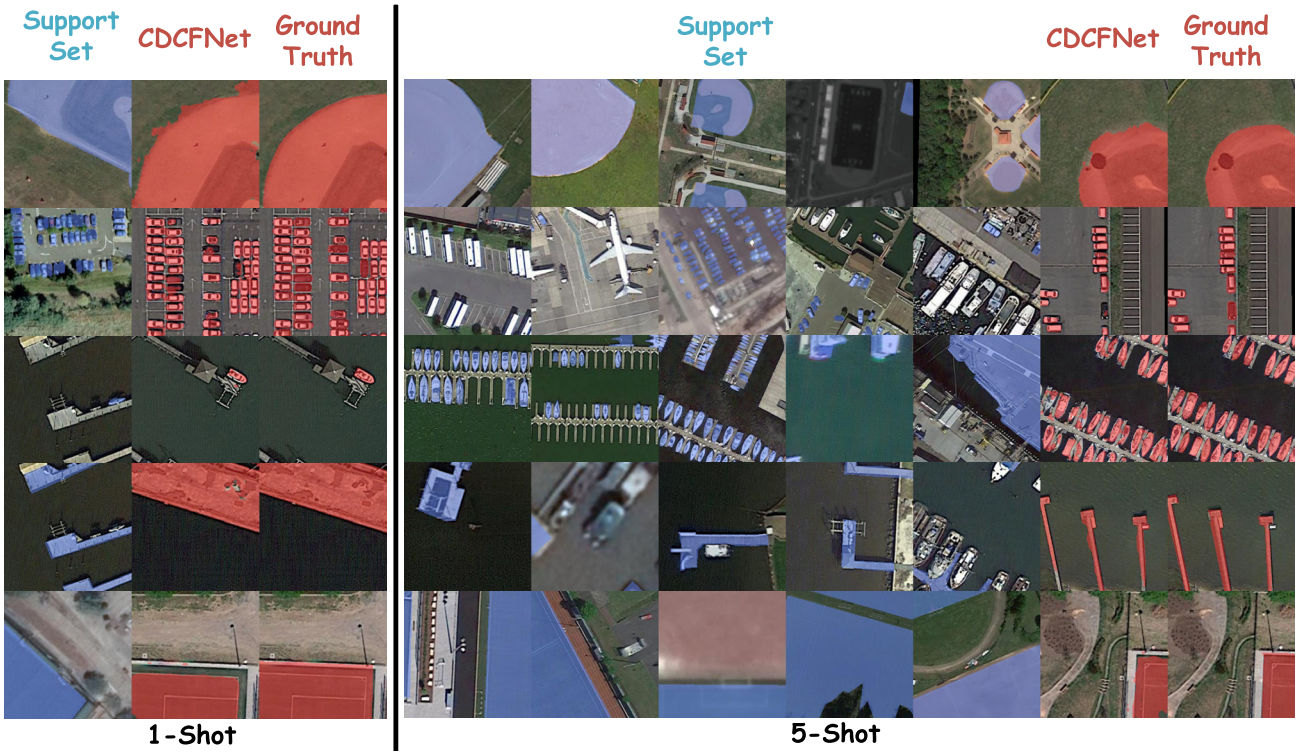


Fig. 10. Visualization results for the DLRSD-4^i $\rightarrow$ iSAID-4^i cross-domain scenario.

proposed framework under heterogeneous data distributions. These results demonstrate that increasing the number of support images facilitates more robust cooperative learning in CDCFNet, and deeper backbone networks enable finer-grained feature extraction.

In addition, Figs. 10 and 11 present several segmentation visualization results of CDCFNet under the one-shot and five-shot settings for the cross-domain tasks DLRSD-4^i $\rightarrow$ iSAID-4^i and iSAID-4^i $\rightarrow$ DLRSD-4^i, respectively. As illustrated in the figures, even with only a single labeled

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TABLE III

OVERALL PERFORMANCE (IN %) COMPARISON ON THE MULTICATEGORY CROSS-DOMAIN FEW-SHOT SEGMENTATION DATASET; THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	DLRSD-4^i→iSAID-4^i				iSAID-4^i→DLRSD-4^i			
		1-Shot		5-Shot		1-Shot		5-Shot	
		mIoU	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**
PFENet [16]	ResNet-50	19.18	50.06	22.49	53.41	19.76	43.57	25.45	47.26
VAT [55]		24.46	49.11	29.10	51.92	23.00	42.97	26.29	45.83
HSNet [11]		25.24	**52.96**	32.88	56.24	26.57	45.92	30.56	48.25
BAM [12]		21.79	51.44	26.27	55.81	21.56	44.78	24.79	39.68
HDMNet [56]		24.67	51.80	31.48	55.90	25.82	45.47	29.76	47.37
R^2Net [18]		22.85	40.51	27.39	43.84	21.70	38.36	24.81	40.52
PATNet [21]		25.79	51.49	36.63	56.71	26.86	46.37	31.03	48.34
CDCFNet		**26.87**	52.49	**37.77**	**57.65**	**27.64**	**47.21**	**31.92**	**49.78**
PFENet [16]	ResNet-101	19.37	50.79	30.67	55.37	21.52	37.04	24.21	40.08
VAT [55]		24.92	50.39	31.24	51.72	23.87	42.63	27.10	45.71
HSNet [11]		25.74	52.85	33.23	56.22	25.48	45.78	31.34	47.94
BAM [12]		22.19	52.41	27.56	**56.32**	23.16	45.87	26.99	41.83
HDMNet [56]		25.72	52.73	32.77	56.19	26.87	**46.83**	30.58	48.51
R^2Net [18]		23.61	38.18	27.86	45.19	22.83	37.19	26.47	39.12
PATNet [21]		27.14	52.07	36.53	54.06	27.35	46.02	31.55	48.11
CDCFNet		**28.85**	**53.21**	**38.72**	54.73	**28.01**	46.70	**32.31**	**48.97**

TABLE IV

OVERALL PERFORMANCE (IN %) COMPARISON ON THE MULTICATEGORY CROSS-DOMAIN FEW-SHOT SEGMENTATION DATASET WITH AND WITHOUT CDCFNET†, THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	DLRSD-4^i→iSAID-4^i				iSAID-4^i→DLRSD-4^i			
		1-Shot		5-Shot		1-Shot		5-Shot	
		mIoU	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**
VAT [55]	ResNet-50	24.46	49.11	29.10	51.92	23.00	42.97	26.29	45.83
VAT [55]+CDCFNet†		25.44 (+0.98)	50.21 (+1.10)	30.29 **(+1.19)**	52.63 (+0.71)	23.89 (+0.89)	44.05 **(+1.08)**	27.24 **(+0.95)**	46.78 **(+0.95)**
HSNet [11]		25.24	52.96	32.88	56.24	26.57	**45.92**	30.56	**48.25**
HSNet [11]+CDCFNet†		**26.05** (+0.81)	**53.01** (+0.05)	**33.57** (+0.69)	**57.21** (+0.97)	**26.92** (+0.35)	45.83 (-0.09)	**30.84** (+0.28)	48.15 (-0.10)
R^2Net [18]		22.85	40.51	27.39	43.84	21.70	38.36	24.81	40.52
R^2Net [18]+CDCFNet†		24.01 **(+1.16)**	42.13 **(+1.62)**	28.41 (+1.02)	45.22 **(+1.38)**	22.76 **(+1.06)**	38.45 (+0.09)	25.61 (+0.80)	41.15 (+0.63)
VAT [55]	ResNet-101	24.92	50.39	31.24	51.72	23.87	42.63	27.10	45.71
VAT [55]+CDCFNet†		26.14 (+1.22)	51.83 (+1.44)	32.35 (+1.11)	53.47 (+1.75)	24.64 (+0.77)	43.89 **(+1.26)**	28.02 (+0.92)	46.57 (+0.86)
HSNet [11]		25.74	52.85	33.23	56.22	27.48	45.78	31.34	47.94
HSNet [11]+CDCFNet†		**27.94** **(+2.20)**	**53.49** (+0.64)	**36.31** **(+3.08)**	**56.81** (+0.59)	**27.93** (+0.45)	**46.85** (+1.07)	**31.79** (+0.45)	**48.64** (+0.70)
R^2Net [18]		23.61	38.18	27.86	45.19	22.83	37.19	26.47	39.12
R^2Net [18]+CDCFNet†		25.06 (+1.45)	40.53 **(+2.35)**	29.65 (+1.79)	48.28 **(+3.09)**	23.90 **(+1.07)**	38.29 (+1.10)	28.73 **(+2.26)**	40.31 **(+1.19)**

support image (one-shot), the model is capable of producing satisfactory segmentation results for the target regions. When the number of labeled support images increases to five (five-shot), the model achieves more precise delineation of object boundaries and semantic regions, which demonstrates enhanced segmentation accuracy and consistency.

D. Ablation Studies

In order to verify the effectiveness of the proposed method in improving performance and its general applicability across different algorithms, three representative few-shot segmentation models (VAT [55], HSNet [11], and R^2Net [18]) are selected to be integrated with the proposed cross-domain cooperative attention and GAF-PTM modules, which form the enhanced version denoted as "+ CDCFNet†." Both the enhanced and the original methods are evaluated on the multicategory cross-domain few-shot segmentation task, with the results summarized in Table IV.

As shown in the table, CDCFNet† consistently achieves varying degrees of improvement across different methods,

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TABLE V

OVERALL PERFORMANCE (IN %) COMPARISON ON THE SINGLE-CATEGORY CROSS-DOMAIN FEW-SHOT SEGMENTATION DATASET; THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	Inria→CHN6-CUG				CHN6-CUG→Inria			
		1-Shot		5-Shot		1-Shot		5-Shot	
		mIoU	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**	**mIoU**	**FB-IoU**
PFENet [16]	ResNet-50	24.21	40.13	26.88	41.35	8.07	29.57	10.63	32.58
VAT [55]		27.73	49.03	31.81	55.30	12.24	38.23	12.67	37.96
HSNet [11]		28.85	49.51	32.10	54.92	11.86	38.97	14.81	38.52
BAM [12]		29.14	51.13	32.26	54.18	10.44	38.92	13.29	39.34
HDMNet [56]		28.79	50.11	31.63	54.71	11.01	38.36	13.03	39.83
R^2Net [18]		30.05	52.21	33.59	55.72	12.89	**39.36**	14.21	**41.25**
MS^2A^2Net [19]		28.46	50.22	32.39	55.84	12.70	38.47	13.17	38.65
CDCFNet		**31.19**	**52.48**	**35.30**	**56.37**	**13.92**	39.06	**15.32**	40.35
PFENet [16]	ResNet-101	27.41	43.36	30.05	45.64	11.35	33.74	14.16	35.17
VAT [55]		29.84	51.72	33.13	55.16	14.67	**41.23**	16.20	41.41
HSNet [11]		29.42	51.50	32.43	55.37	14.76	40.13	15.86	41.04
BAM [12]		32.22	53.13	35.42	55.15	13.54	40.87	15.94	42.53
HDMNet [56]		31.24	53.46	34.84	56.53	15.03	40.27	17.47	**42.69**
R^2Net [18]		32.78	**55.18**	**37.16**	57.28	15.16	41.08	**18.07**	42.01
MS^2A^2Net [19]		30.35	52.18	33.42	55.96	14.95	41.10	16.92	41.81
CDCFNet		**32.84**	54.37	36.44	**57.93**	**15.27**	40.30	17.67	42.15

TABLE VI

OVERALL PERFORMANCE (IN %) COMPARISON ON THE HETEROGENEOUS-CATEGORY CROSS-DOMAIN FEW-SHOT SEGMENTATION DATASET; THE BEST RESULT IS INDICATED IN BOLD

Category	1-Shot							5-Shot						
	VAT	HSNet	BAM	HDMNet	R^2Net	MS^2A^2Net	**CDCFNet**	VAT	HSNet	BAM	HDMNet	R^2Net	MS^2A^2Net	**CDCFNet**
Helicopter	13.80	8.33	11.01	11.43	**14.71**	12.23	14.56	15.48	10.25	12.45	13.43	16.92	13.19	**17.24**
Small Vehicle	10.84	8.77	10.97	11.12	12.67	9.60	**13.31**	14.16	12.63	14.50	14.65	17.41	13.39	**17.91**
Storage Tank	44.00	39.22	39.96	42.58	42.91	43.38	**44.25**	47.29	42.38	42.84	45.45	45.02	47.18	**48.57**
Tennis Court	52.25	49.61	49.21	49.87	56.32	56.45	**57.15**	63.06	59.03	58.25	58.03	**68.05**	66.81	67.77
Basketball Court	24.73	21.13	22.30	23.65	**25.94**	25.83	25.88	26.89	23.67	26.93	28.57	30.53	29.37	**30.55**
Bridge	9.72	5.86	7.61	10.83	11.48	8.37	**12.77**	12.98	9.65	11.24	14.46	16.80	11.41	**17.70**
Large Vehicle	37.44	31.41	32.83	35.92	37.21	37.16	**39.25**	45.51	40.19	42.96	45.14	46.89	46.79	**48.62**
Soccer Field	41.25	39.63	41.34	41.67	43.52	43.22	**43.73**	47.89	45.20	45.67	47.96	49.93	48.89	**51.67**
Ship	36.07	34.59	35.76	35.66	36.30	37.46	**39.80**	40.05	37.22	38.38	38.45	38.78	39.51	**43.42**
Ground Track Field	16.16	13.89	14.69	14.66	15.66	14.37	**17.43**	24.06	19.23	22.94	20.89	21.16	20.75	**24.68**
Roundabout	16.12	14.03	12.24	13.12	17.50	17.15	**18.30**	21.56	18.56	18.60	18.29	**23.76**	22.94	23.35
Swimming Pool	13.83	12.39	11.40	13.67	16.00	14.71	**16.23**	21.25	20.75	21.13	24.53	25.06	24.84	**26.09**
Harbor	25.28	20.58	23.17	22.51	23.80	25.85	**26.68**	32.20	25.87	29.02	27.61	30.33	**33.18**	32.50
Baseball Diamond	24.26	23.97	21.24	22.05	25.40	23.12	**27.59**	34.48	34.02	32.30	33.32	35.83	34.48	**39.57**
Plane	38.19	37.21	38.74	40.00	39.97	40.57	**41.43**	43.56	46.20	46.34	45.38	47.64	47.61	**50.78**
mIoU	26.93	23.98	24.83	25.98	27.96	27.30	**29.29**	32.69	29.66	30.90	32.74	34.27	33.23	**36.03**

which confirms the effectiveness of the proposed approach and adaptability to diverse methods. In addition, the performance improvements become more substantial under the five-shot setting, indicating that CDCFNet† can effectively leverage the information provided by additional support samples. Under identical experimental conditions, the performance gains achieved with the ResNet-101 backbone are consistently greater than those with ResNet-50, which means that deeper backbone architectures further enhance the feature representation capacity and cross-domain collaboration effectiveness of CDCFNet†.

E. Extension Experiments

To further validate the cross-domain segmentation performance of the proposed method in scenarios with minimal data distribution overlap, extended experiments are conducted on the single-category and heterogeneous-category cross-domain few-shot segmentation datasets.

Table V presents the comparison results of different few-shot segmentation algorithms on the single-category cross-domain dataset with ResNet-50 and ResNet-101 backbones. As shown in the table, CDCFNet demonstrates a clear advantage in heterogeneous cross-domain few-shot segmentation tasks. With ResNet-50 as the backbone, CDCFNet achieves an mIoU of 31.19 in the building-to-road transfer task with the one-shot setting, which surpasses the best baseline R^2Net by 1.14 in mIoU and 0.27 in FB-IoU. In the five-shot setting, the improvements reach 1.71 and 0.65, respectively. For the reverse transfer (road → building), CDCFNet still outperforms R^2Net in mIoU, though the FB-IoU is slightly

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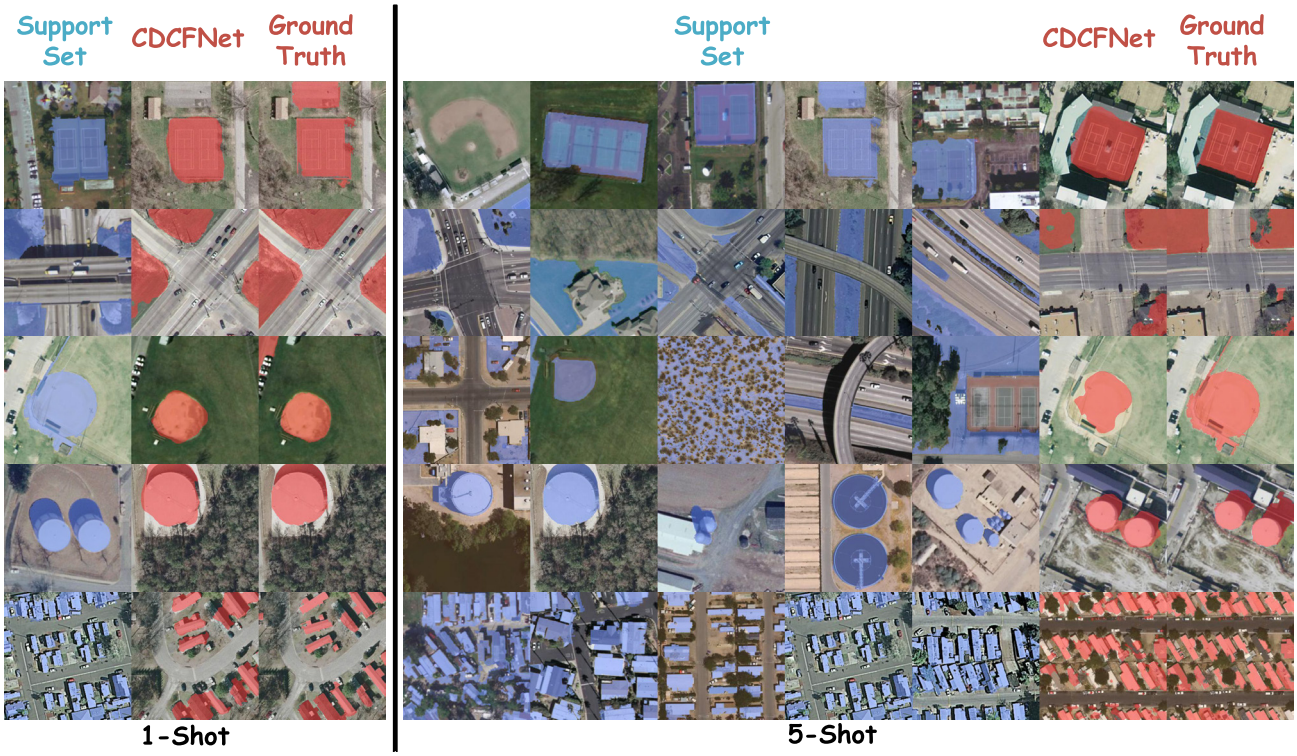


Fig. 11. Visualization results for the iSAID-4^i → DLRSD-4^i cross-domain scenario.

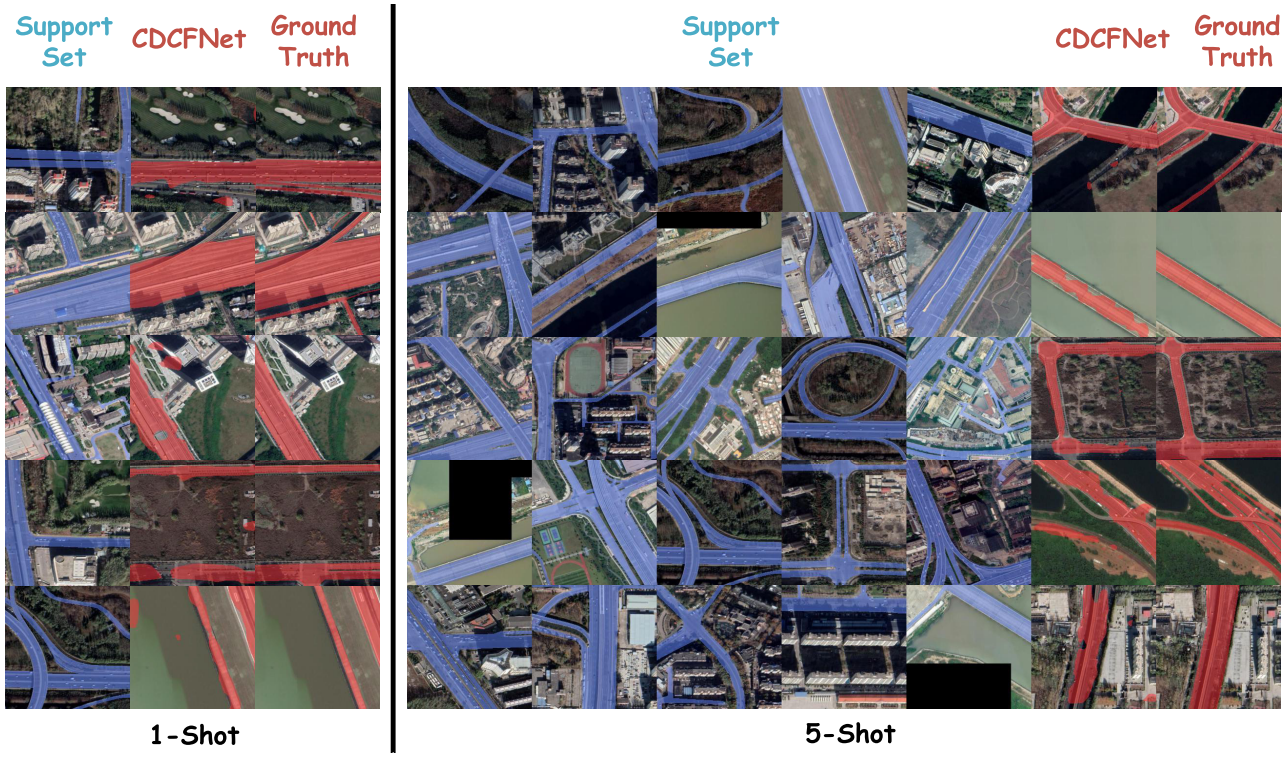


Fig. 12. Visualization results for the Inria (building) → CHN6-CUG (road) cross-domain scenario.

lower. It is worth noting that in cross-domain segmentation tasks, mIoU is generally a more representative metric of model generalization than FB-IoU. When ResNet-101 is adopted as the backbone, CDCFNet achieves further improvements, which benefit from its ability to capture fine-grained semantic cues such as building edges and road textures, and the effectiveness of multiscale gating mechanisms.

Another notable result is that the road-to-building transfer consistently underperforms the building-to-road transfer. This disparity arises from intrinsic feature differences.

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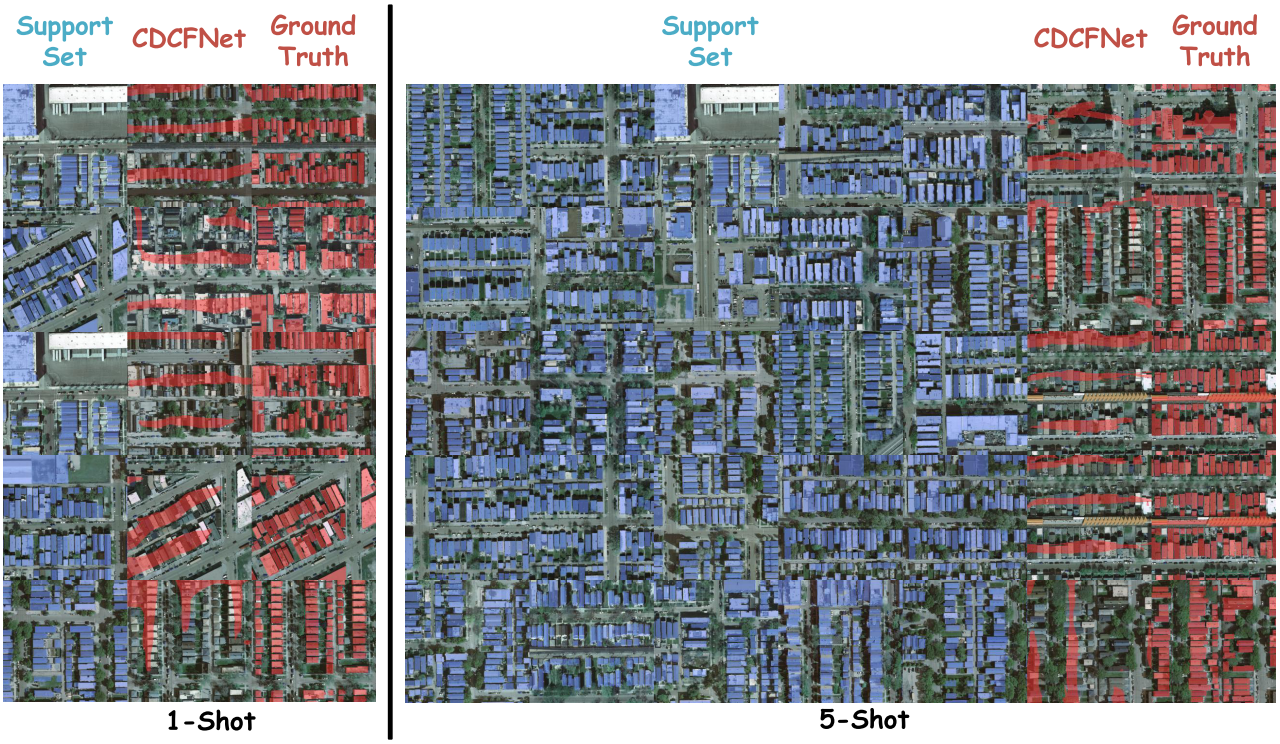


Fig. 13. Visualization results for the CHN6-CUG (road) → Inria (building) cross-domain scenario.

Buildings exhibit high semantic richness and geometric diversity, which provides transferable knowledge such as structural regularity, while roads are characterized by linear and low-texture patterns with limited diversity. In the building-to-road transfer, CDCFNet leverages gated mechanisms to extract general structural features from buildings and map them to simpler road patterns. Conversely, in reverse transfer, road features struggle to cover building diversity, which hinders the ability to learn discriminative semantics. Although the deeper ResNet-101 backbone partially mitigates this issue, its five-shot mIoU (17.67) remains substantially lower than that of the building-to-road transfer task (36.44), revealing the inherent link between source-domain feature richness and target-domain complexity.

Fig. 12 shows the visualization results of building-to-road transferring. As seen in the figure, even though the training set only contained building samples, the model still achieves strong performance on road segmentation. With only one annotated image as reference (one-shot), CDCFNet can roughly segment the main road regions. When five reference images are provided (five-shot), the segmentation is more precise, and the model successfully captures complex road structures. Specifically, in the second visualization example under the one-shot setting, the model accurately segments the main road but fails to fully capture the branching paths. In contrast, as shown in the first visualization under the five-shot setting, CDCFNet not only delineates the main road but also accurately segments both branch roads. This demonstrates that increasing the number of reference samples enables the model to learn more diverse and detailed road representations. In addition, as illustrated in the fifth visualization of the one-shot setting, although the background between two adjacent roads appears visually similar to the road itself, CDCFNet still successfully distinguishes the foreground boundaries, which is attributed to the strong ability to capture fine-grained edge features.

Fig. 13 illustrates the visualization results of the cross-category transfer experiment from road to building. It can be seen that when only one annotated image is provided as a reference, CDCFNet can roughly localize parts of the building regions but fails to accurately segment the complete targets. Even with five reference images, the segmentation quality shows no substantial improvement. For example, in the first visualization under the one-shot setting, the model captures the general location of the buildings but misses fine details and boundary structures. Similarly, the first five-shot result indicates that increasing the number of support samples does not significantly enhance the ability of the model to capture architectural details. These results are attributed to the limited structural diversity of road features, which leads to insufficient transferable information and constrains the ability to adapt to complex categories such as buildings. This also reflects an inherent asymmetry in cross-category transfer, where features learned from structurally simple categories are less capable of generalizing to more complex structures. Furthermore, in both one-shot and five-shot visualizations, the predicted masks appear as continuous regions, making it difficult to separate individual buildings. Due to the strong linear continuity of source-domain features (roads), CDCFNet suffers from overfitting during domain adaptation, which in turn weakens its ability to disentangle fine-grained semantic details such as the boundaries between adjacent buildings.

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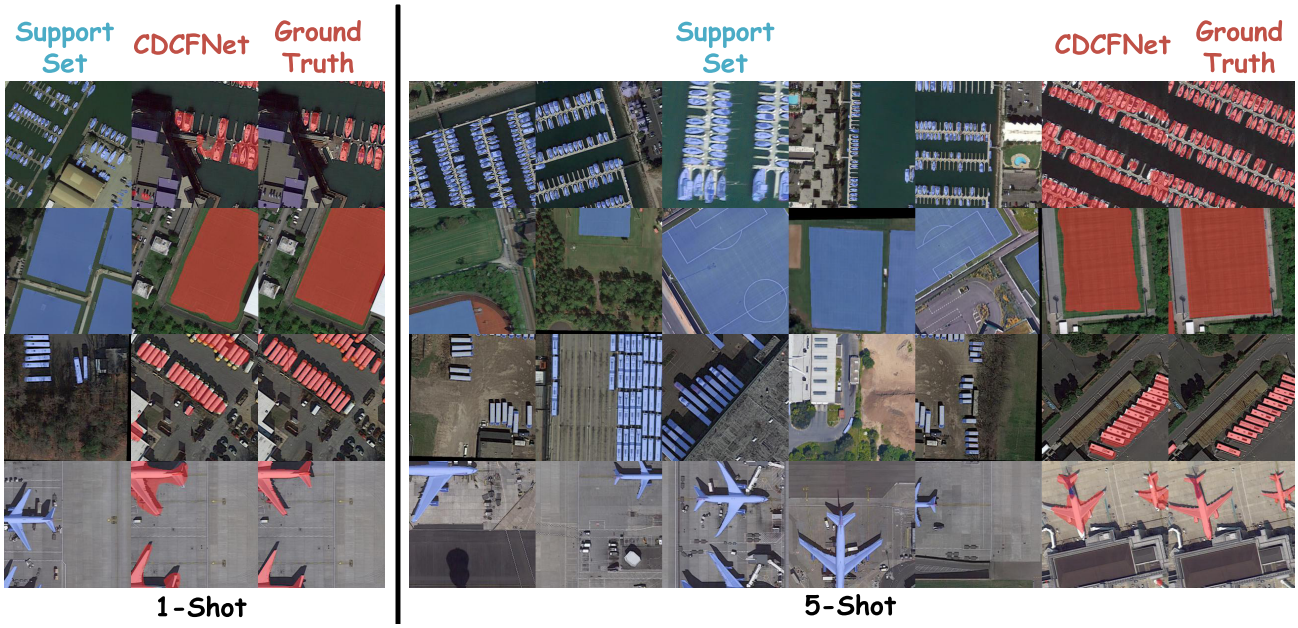


Fig. 14. Visualization results for the Inria (building) → iSAID-5^i cross-domain scenario.

To provide a more detailed comparison of heterogeneous-category cross-domain few-shot segmentation performance, Table VI presents the experimental results of various methods using ResNet101 as the backbone, where each method transfers from the Inria building category to each target category in the iSAID-5^i dataset. Throughout all previous experiments, PFENet consistently exhibited substantially lower performance than other methods. Given that the iSAID-5^i dataset contains a large number of categories, PFENet is excluded from subsequent experiments for clarity and table compactness.

As shown in Table VI, across 15 categories, CDCFNet achieves the highest IoU in 13 categories for one-shot experiments and in 12 categories for five-shot experiments. Compared with the next-optimal method R^2Net, CDCFNet achieves improvements of 1.33 in the one-shot case and 1.76 in the five-shot case, which demonstrates that a deeper backbone network significantly enhances the collaborative feature processing and cross-domain adaptability, and leads to superior segmentation performance.

Fig. 14 presents partial visualization results of the heterogeneous-category cross-domain few-shot segmentation from the building domain to the iSAID-5^i data. It can be seen that although the training set contains only building samples, CDCFNet achieves satisfactory segmentation performance across various target categories. In the one-shot setting, CDCFNet can already segment most target objects accurately. When the number of reference images increases to five, the segmentation results become noticeably more precise. For instance, in the second five-shot visualization, CDCFNet successfully delineates the target region even when it appears at the image edge and is partially occluded, which verifies the strong robustness of the model. In addition, in both the one-shot and five-shot results, CDCFNet accurately segments scenes with multiple objects, with each object precisely separated from others. These results confirm that CDCFNet is capable of capturing fine-grained structural details with high stability and precision.

V. Conclusion

This article proposes a CDCFNet to address the challenges of data scarcity, domain discrepancy, and complex category variation in cross-domain few-shot remote sensing imagery segmentation. On the one hand, the proposed method integrates cross-domain cooperative attention to achieve adaptive feature alignment and effectively integrate spatial positional and contextual information, which enhances feature consistency and semantic alignment across domains. On the other hand, a gated adaptive feature transformation is designed to perform hierarchical modeling and domain-invariant mapping of source and target domain features. This effectively separates domain-related features from transferable features and mitigates the distribution shift. Comprehensive experiments on multiple remote sensing datasets demonstrate that the proposed method achieves consistent improvements across various few-shot cross-domain segmentation tasks, which validates its effectiveness and generalization capability.

However, the generalization capability of the proposed method under low-diversity source domains still needs to be improved. Therefore, our future work will explore incorporating adversarial domain adaptation and hierarchical attention mechanisms to enhance robustness and generalization performance in complex cross-domain scenarios. In addition, we plan to extend the framework to cross-modality remote sensing segmentation by incorporating modality-invariant alignment mechanisms.

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